



شبكات الحاسب

8:30-9:30

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د/ابرام كمال

نظم معلومات حيوية

Faculty of Computers & Information, Assiut University

Course code: IT351

3rd Level

Midterm Exam

Duration: One hour

علي هاشم محروس عبدالحافظ

2

رقم الجلوس *

1620195167

3

المستوي *

- ☐ الاول
- ☐ الثاني
- ☐ الثالث
- ☐ الرابع 2013
- ☐ الرابع 2014
- ☐ الرابع 2015
- ☐ الرابع 2016

☐ الرابع 2017

☐ الرابع 2018

☒ الرابع 2019

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الكود (قد تمت مراجعة بيانات الطالب ورقم الجلوس) *

KAIW

📝 Will be reviewed

8

How many sockets are there in a TCP server communicating with 13 clients concurrently?



- ☐ 15
- ☐ None of the options
- ☐ 13
- ☐ 12
- ☒ 14

📝 Will be reviewed

9

A source and a destination are separated by 3×10^5 kilometers and are connected by a direct link of 2 Kbps. The propagation speed over the link is 2×10^8 meters/second. The source sends 100 packets to the destination using RDT 2.2. Each packet is of 2×10^3 bits long.

Suppose ACK packets are of negligible size and transmission channel is perfectly reliable. What is the throughput (in bps) of transmission?

- ☐ 100
- ☒ 500
- ☐ 400
- ☐ None of the options
- ☐ 200

 **Will be reviewed**

10

Which of the following statements is true about TCP and not true about UDP?



- ☐ Provides timing service.
- ☒ Provides reliable transport service.
- ☐ Provides security service.
- ☐ Provides minimum throughput guarantee service.

 **Will be reviewed**

11

Top-level domain (TLD) servers are organization's own DNS server(s), providing authoritative hostname to IP mappings for organization's named hosts.



☐ No

☒ Yes

 **Will be reviewed**

12

Which of the following layers doesn't appear in the Internet Protocol Stack?



☐ Physical layer

☒ Session layer

☐ Application layer

☐ None of the options

- ☐ Network layer
- ☐ Transport layer

✍ **Will be reviewed**

13

Consider the following pseudo-code for rdt 3.0 receiver.

Here, **seq** is the expected sequence number (either 0 or 1) by the receiver. The meaning of the rest function calls and variables are the same as those in FSM diagram. We say that the receiver waits forever, if it is blocked at the while loop, waiting to receive a packet that will never arrive. We say that the sender loops forever, if it repeatedly retransmits the same packet over and over again. Which of the following statements is TRUE? 

- ☒ A single loss ACK packet is sufficient to cause the sender to loop forever.
- ☐ A single corrupted data packet is sufficient to cause the sender to loop forever.
- ☐ A single corrupted data packet is sufficient to cause the receiver to wait forever.
- ☐ A single premature timeout is sufficient to cause the sender to loop forever.

✍ **Will be reviewed**

14

Which of the following statements is true about circuit switching? 

- ☐ The path between a source and a destination is not fixed
- ☐ Packet transmission can use the full link bandwidth
- ☐ None of the options
- ☒ Resources are pre-allocated regardless of demand
- ☐ All of the options

 **Will be reviewed**

15

A packet switch receives a packet and determines the outbound link to which the packet should be forwarded. When the packet arrives, x bits of the currently-being-transmitted packet have been transmitted on this outbound link and n other packets are waiting to be transmitted. Packets are transmitted continuously in order of arrival. Suppose all packets are L bits and the link rate is R bps.

What is the queuing delay?



☐ $(nL + (L - x)) / R$

☐ nL / R

☐ $n(L - x) / R$

☒ $nL + (L - x) / R$

☐ $(nL + (x - L)) / R$

 **Will be reviewed**

16

In Go-back-N protocol, what are the size of the respective sender and receiver buffers required for a window size of N? 

☐ N; N

☒ N; 1

☐ 1; 1

☐ N - 1; N - 1

☐ N - 1; 1

 **Will be reviewed**

17

Which of the following statements about DNS is TRUE?



- ☐ None of the options
- ☐ Every Web server must have a canonical name.
- ☒ Local DNS server sometimes may provide out-of-date hostname-to-IP-address mapping.
- ☐ DNS server listens to TCP port 53.
- ☐ TLD servers store all the hostname to IP mappings of the Internet.

 **Will be reviewed**

18

A Web server stores a webpage that comprises a base HTML file and 2 images referenced by the base HTML file. The HTML file is 100 bytes and each image is 200 bytes. A client is connected to the Web server through a direct link of 1 Mbps. Propagation delay between the Web server and the client is 50 milliseconds.

The client downloads the webpage using persistent HTTP but without pipelining (i.e. the next HTTP request is sent after the response for the previous HTTP request is received). Assume

HTTP header, TCP header and ACK packets are of negligible size, transmission channel is perfectly reliable, time to establish and close TCP connection can be ignored. How long (in milliseconds) does it take for the client to download the entire webpage from the Web server?



- ☐ 604
- ☒ 204
- ☐ 402.4
- ☐ 404

 **Will be reviewed**

19

HTTP protocol keeps state information at the server side about past client requests.



- ☒ True
- ☐ False

 **Will be reviewed**

20

Which of the following statements is true about packet switching?



- ☐ In packet switching, dedicated allocation is done between nodes.
- ☐ In packet switching, time is divided into "slots" between the nodes.
- ☒ In packet switching, bandwidth is divided into "pieces" between the nodes.
- ☐ In packet switching, resources are reserved for nodes.
- ☐ None of the options.

 **Will be reviewed**

21

In Non-persistent HTTP, the client closes the TCP connection after fetching one object (one pair of HTTP request and HTTP response)



- ☐ False
- ☒ True

 **Will be reviewed**

22

Can HTTP response message contains an empty body? If yes, When can this happen?



- ☐ No, an HTTP response message can never have an empty body.
- ☐ Yes, if the requested object is recently modified.
- ☒ Yes, if the requested object is very small in size.
- ☐ Yes, if the requested object has been moved to a new location.

 **Will be reviewed**

23

In circuit switching, circuit segment is idle if not used by call. 

- ☒ True
- ☐ False

 **Will be reviewed**

24

A browser requests to download index.html from a Web server. The HTTP response header received by the browser is shown below.

Which of the following statements is TRUE?



- ☐ Suppose TCP header is 20 bytes, the length of the TCP segment containing the HTTP response is 1276 bytes.
- ☐ In the HTTP request, browser has requested for a non-persistent connection.
- ☐ None of the options.
- ☐ The index.html file received by the browser may have been corrupted during transmission.
- ☒ The Web server uses a cookie to keep the state information of the client.

 **Will be reviewed**

25

Assume a sender and a receiver connected via one packet-switch. The sender sends a message of size 24 KBytes using packets of length 1000 Bytes. The bit rates are 2 kbps and 5 kbps for the first and the second link, respectively. Using store and forward, what is the end-to-end

delay (in seconds)? You may ignore the propagation delay!



- ☐ 16800
- ☒ 134.4
- ☐ 16.8
- ☐ None of the options
- ☐ 0.1008

 **Will be reviewed**

26

End-to-end delay is the time taken for a packet to travel from source to destination. It consists of which of the following delays?



- ☐ Propagation delay.
- ☐ Queuing delay.
- ☐ Processing delay.

- ☒ All the options.
- ☐ Transmission delay.
- ☐ None of the options.

 **Will be reviewed**

27

Which of the following statements is true?



- ☒ The Internet uses packet switching and traditional telephone networks uses circuit switching.
- ☐ Performance metrics such as end-to-end delay and throughput can be guaranteed in the Internet, if TCP is chosen as the transport layer protocol.
- ☐ None of the options
- ☐ In the Internet, packets from the same source always take the same path to reach destination.
- ☐ Both the Internet and traditional telephone networks use packet-switching.

 **Will be reviewed**

28

Which of the following HTTP protocols does the given figure represent to fetch a web page with 2 reference objects?



- ☒ Persistent HTTP with pipelining.
- ☐ Non-persistent HTTP.
- ☐ Persistent HTTP.
- ☐ Non-persistent HTTP with parallel connection.

 **Will be reviewed**

29

Which of the following statements about socket programming with TCP is true?



- ☐ Server cannot send data to client after a client initiates TCP connection to the server.
- ☐ Client explicitly attaches destination IP address and port number to every packet
- ☒ Server extracts sender IP address and port number from the received packet.

☐ None of the above

 **Will be reviewed**

30

Which of the following statements is true about Client-Server Architecture?



- ☐ In Client-Server Architecture, the server typically requests service from the client while the client provides requested service to server.
- ☒ In Client-Server Architecture, the server waits for incoming requests while the client Initiates contact with server.
- ☐ In Client-Server Architecture, the client waits for incoming requests while the server Initiates contact with server.
- ☐ All of the options.
- ☐ None of the options.

 **Will be reviewed**

31

A Selective Repeat sender just receives an ACK packet with ACK number 7. This ACK number falls within sender window which has the window size 3. Every data packet embeds a k-bit sequence number field (k is a constant unknown to you).

Which of the following definitely CANNOT be the sequence number of the next packet transmitted by the sender? 

☒ 3

☐ 0

☐ 10

☐ 2

☐ 6

 **Will be reviewed**

32

Using Conditional GET in HTTP Protocol, the number of RTT is reduced if the object is not modified since the date of the cached copy!



☒ False

☐ True.

 **Will be reviewed**

33

Consider the transmission between a UDP sender and a UDP receiver. Which of the following will never happen? You may assume that the application riding on UDP doesn't implement any reliability mechanisms. 

- ☒ UDP receiver receives duplicate packets from UDP sender.
- ☐ None of the above
- ☐ UDP receiver receives out-of-order packets from UDP sender.
- ☐ UDP receiver fails to receive any packet from UDP sender.
- ☐ UDP receiver receives corrupted packets from UDP sender but fails to detect bit errors.

 **Will be reviewed**

34

What is the checksum (1's complement of the sum) of the following 3 bytes?

11110100

10010011

10011001 

☒ 11011101

☐ 11011110

☐ 00100010

☐ 11011111

 **Will be reviewed**

35

Which of the following is present in both HTTP request line and status line?



☒ Status code

☐ None of the options

☐ URL

☐ HTTP version number

☐ Request method

 **Will be reviewed**

36

Consider the following Python code snippet.

Suppose no exception is raised when the above code snippet is executed, which of the following statements is TRUE? 

- ☐ s2 listens to a random port number assigned by operation system
- ☐ s1 is a UDP socket
- ☐ None of the options.
- ☐ Server uses s1 to transmit application data to client
- ☒ s2 listens to port 8080

 **Will be reviewed**

37

Consider a sender and a receiver communicating using Selective Repeat protocol. Every packet embeds a 4-bit sequence number field. Sender window size is 4. None of the packets

shown in the following figure are corrupted packets. However, the third data packet and the second ACK packet are lost.

What is the sender window over the sequence number space at time t? 

- ☐ 13, 14, 0, 1
- ☐ 12, 13, 14, 15
- ☒ 13, 14, 15, 0
- ☐ 13, 14, 15, 16



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